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**A Project Stage-I Report**

On

# SUSTAINABLE HYDROPONIC FARMING - INTELLIGENT AUTOMATION AND MONITORING

*Submitted in the partial fulfillment of the requirements of*

**Bachelor of Engineering**

**in**

**Electronics and Telecommunication**

By

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Under the guidance of

**Prof. R.R. Itkarkar**

**Department of Electronics and Telecommunication Engineering**

**All India Shri Shivaji Memorial Society's College of Engineering, Pune-01**

**Affiliated to SPPU Pune**

**A.Y 2025-26**

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## CERTIFICATE

This is to certify that the Project titled "**Sustainable hydroponic farming - intelligent automation and monitoring"** Submitted by,

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is a bonafide record of the Project Stage-I work carried out by them towards the partial fulfillment of the requirements of Savitribai Phule Pune University, for the award of Bachelor of Electronics and Telecommunication Engineering under my supervision and guidance.

Dr. R.R.Itkarkar
Guide and Project Coordinator

Dr. S.B. Dhonde
HOD-E & TC

Date:

Place**: Pune**

This report has been examined as per Savitribai Phule Pune University requirements at **AISSMS COE, Pune-01** on ________________

**Signature:**

**Name of Internal Examiner**

**Signature:**

**Name of External Examiner**

# ACKNOWLEDGEMENT

We take this opportunity to express our deep sense of gratitude towards our guide and our project coordinator Prof. R R Itkarkar under whose guidance we had the privilege to work on this project. We also express our gratitude towards our project co-ordinator Prof. R R Itkarkar for her kind support and co-operation.

We are grateful to Dr S B Dhonde Head of E & TC Engineering Department and all the teaching and non-teaching staff members of E& TC Engineering Department for their direct or indirect help in completion of the Stage-I of project.

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# 1. INTRODUCTION

In the era of rising global population, diminishing arable land, and increasingly unpredictable climatic conditions, traditional soil-based agriculture faces critical challenges. Conventional farming methods are struggling to keep up due to their extensive water usage, soil degradation, and vulnerability to climatic fluctuations.To address food security sustainably, innovative farming techniques are essential. Hydroponics is a method of growing plants without soil, using nutrient-rich water solutions and has emerged as a promising alternative. By delivering tailored nutrition and optimizing environmental factors such as temperature and humidity, hydroponic systems offer enhanced crop yields with drastically reduced land and water requirements.

Unlike conventional agriculture which depends heavily on fertile soil, hydroponics provides controlled conditions for plant growth. This soil-free approach eliminates many traditional risks like soil-borne diseases and pests, improves nutrient uptake efficiency, and facilitates year-round cultivation. Numerous crops ranging from leafy greens and herbs to fruit-bearing plants like tomatoes and peppers can be grown effectively through hydroponics. This makes it highly suitable for urban farming and areas with poor soil quality.

Successful hydroponic farming critically depends on precise control of environmental parameters such as nutrient concentration, pH level, temperature, and humidity. Maintaining these conditions in optimal ranges ensures healthy plant growth, maximized yield, and improved quality. However, manual monitoring and adjustment of such parameters can be labor-intensive and prone to errors, especially at larger scales.To truly harness the benefits of hydroponics, integrating smart technologies has become necessary. Internet of Things (IoT) sensor networks enable continuous monitoring of critical growth parameters such as pH, nutrient concentration (TDS and EC), temperature, humidity, water flow, and water levels. These data facilitate automated control of nutrient dosing, irrigation, climate regulation, and alerts which minimize human intervention, reduce errors, and optimize resource use.

This project focuses on developing a smart hydroponic farming system that uses a powerful ESP32-S3 microcontroller integrated with multiple environmental sensors. The system communicates wirelessly with a mobile app platform (Blynk) to provide real-time monitoring, remote control, and notifications. Through simulation, component selection, and hardware design, the project aims to create an affordable, scalable prototype that promotes sustainable agricultural practices. By combining hydroponics with IoT innovation, this project addresses urgent concerns in modern agriculture by maximizing productivity on limited land, conserving resources, reducing environmental impact, and empowering growers with precision tools.

## 2. PROBLEM STATEMENT

To design and develop a fully automated hydroponic farming system using ESP32 and IoT-based wireless monitoring.

## OBJECTIVE

- Design the system to monitor various parameters for hydroponic farming area such as pH , EC (electrical conductivity), temperature, humidity, control nutrient dosing, control pH level, automate fan and humidifier.
- Integrate the above parameters with the controller.
- Future analysis and monitor through cloud .

# 3. LITERATURE SURVEY

[1] This paper presents an IoT-integrated hydroponic vertical farming system using sensors like DHT11, LDR, soil moisture, and ESP8266 to monitor and control irrigation, lighting, and environmental conditions remotely via ThingSpeak and Blynk app. It emphasizes optimized plant growth, resource efficiency, and reduced human intervention. The research gap lies in high costs and complexity hindering wider adoption and the need for further testing to enhance system effectiveness.

[2] The work proposes a real-time monitoring system using sensors for pH, temperature, humidity, light intensity, water level, and TDS, with cloud-based data storage and APIs for remote monitoring and alerts. It implements automatic adjustment of water, nutrients, and environmental factors to improve crop yield and reduce human effort. Challenges include managing heterogeneous wireless communication systems (WiFi and ZigBee) and balancing technical complexity and cost.

[3] This study integrates machine learning (Random Forest, Neural Networks) with IoT to dynamically optimize nutrient delivery (NPK) in hydroponics based on sensor data and historical patterns. It automates nutrient management and environmental controls to improve yield and resource use, validated experimentally. The research gap involves extending adaptive nutrient optimization models to diverse crops and environments, enhancing system scalability and autonomy.

[4] The paper proposes a web-based platform combining IoT environmental monitoring (temperature, humidity, pH, nutrient levels) with Random Forest ML algorithms for real-time control of nutrient delivery, lighting, humidity, and irrigation schedules. It provides modules for crop and task management and market analysis to boost productivity and profitability. Research gaps include incorporating renewable energy sources and expanding multi-crop management features for enhanced sustainability.

[5] This system designs an IoT device integrating multiple sensors (pH, EC, temperature, humidity, water level, color sensors) and a microcontroller (Arduino Uno) linked to Blynk IoT platform for remote monitoring and management of indoor hydroponics. It automates watering and fertilization based on sensor feedback, coupled with alerts. Limitations include vulnerability to power outages and system maintenance requirements.

[6] The paper presents an IoT-based nutrient alert system that continuously monitors pH, temperature, humidity, and nutrient concentrations through sensors linked to a microcontroller and cloud computing. It uses predictive analytics for real-time notification and automated

corrective actions (auto-dosing). The system addresses gaps in conventional setups lacking integration of multi-parameter monitoring, real-time alerts, and predictive AI-based interventions.

[7] This review explores urban vertical farming as a solution for food security with IoT sensor networks deployed for monitoring environmental factors and automation of irrigation, lighting, and nutrient delivery. It highlights the scalability, resource use efficiency, and AI-driven predictive analytics potential. Challenges include high energy demands, economic feasibility, and urban policy integration needed for sustainable wide-scale adoption.

[8] The study designs and tests a vertical farming prototype managed via smart technology to control environment parameters remotely with IoT sensors and Arduino microcontrollers. Results show 29% higher yields vs traditional horizontal farming. It outlines system hardware/software integration, mobile app control, and emphasizes optimizing growing conditions. Research gaps include material choice for safer construction and further automation capabilities.

[9] This review analyses advances in sensor technologies (including hyperspectral imaging), communication, IoT integration, and ML for hydroponics monitoring. It assesses effects on yield, quality, and resource efficiency and highlights challenges in sensor calibration, system maintenance, and Internet dependency. The paper suggests further research in sensor development, AI integration, and scaling of sustainable hydroponic farming.

[10] This paper presents an AI-IoT hybrid system using ESP32 and Random Forest ML for continuous NPK monitoring, water condition control, and automated fertilizer dosing. Remote monitoring via Flask web dashboard and MQTT protocol enables efficient nutrient management and early nutrient deficiency prediction, improving growth by 40%, water savings, and reducing manual labor. Limitations include sensor calibration, power dependency, and potential scale-up challenges.

[11] This implementation involves monitoring key environmental parameters (pH, temperature, humidity, nutrient levels) in a hydroponic setup with real-time IoT and MQTT-based data transmission and mobile alerts. It highlights ease of implementation and benefits like enhanced resource use, sustainability, and increasing crop productivity. Research gaps focus on improving system robustness and expanding functionality.

[12] The research integrates MQTT-based IoT for continuous pH, temperature, and humidity monitoring and control in hydroponics. Through real-time alerts and automated environmental adjustments, it optimizes nutrient availability, plant health, and crop yield. Case studies show

20-30% yield improvements and resource savings. Gaps remain in customization of crop-specific parameters and advanced integration with predictive analytics.

[13] This work proposes an IoT-ML system combining sensor networks (pH, TDS, temperature, humidity, CO2, light) with ML-based image analysis (AlexNet) for freshness detection and anomaly identification. It automates environmental control and nutrient dosing, accessible via web dashboard. Benefits include reduced resource waste (30-40%), improved growth quality, and lower manual intervention. Challenges include balancing computational cost and model accuracy on embedded hardware.

[14] This study implements IoT-enabled horizontal hydroponic system monitoring pH, temperature, humidity, and nutrients with NodeMCU and MQTT for data transmission. It demonstrates remote monitoring and notification for early issue detection and system control. The research highlights the role of IoT in enhancing hydroponic sustainability and scalability. Identified research gaps include further automation and integration of advanced decision support

[15] This research presents an advanced AI framework using an optimized decision tree for automatic nutrient deficiency classification in hydroponic lettuce. The method combines spectral imaging with feature optimization for precise deficiency detection to improve yield and reduce nutrient waste. The gap identified is the limited applicability to other crops and complexity in real-time deployment in commercial setups.

[16] This study uses ARUCO markers and computer vision techniques to improve monitoring of tomato plant leaves in hydroponic systems for disease and growth assessment. It leverages marker-based localization and machine learning for precise leaf detection and health analysis. A gap is noted in scalability for dense canopies and real-time automation under variable lighting conditions.

[17] This paper integrates AI-driven decision support with IoT sensors in a hydroponic system tailored for holy basil crop production. It employs sensor fusion and neural networks for optimizing nutrient delivery, environmental controls, and growth prediction to achieve higher quality harvests. Research gaps include adaptation for other medicinal plants and comprehensive energy optimization.

[18] This field study in hydroponic lettuce examines correlations between nutrient uptake and transpiration using an IoT sensor interface monitoring environmental and plant physiological data. Statistical analysis reveals key factors influencing nutrient absorption for better system control. Gaps lie in generalizing findings across varieties and environmental conditions.

[19] The paper proposes an Industrial IoT communication architecture for agriculture integrating multiple data streams (sensor, image, machine data) over wireless protocols for real-time decision making in hydroponic and agro-industrial setups. Methods include protocol design and data fusion for system reliability and latency reduction. Gaps include deployment challenges in remote areas with limited connectivity.

[20] This comprehensive review analyzes IoT applications in soilless agriculture including hydroponics and aeroponics, covering sensor systems, data analytics, automation, and remote monitoring for sustainable farming. It outlines key opportunities for yield optimization and resource efficiency alongside challenges like data security, energy management, and economic feasibility for farmers. It highlights gaps in standardization and integration with broader smart farming ecosystems.

## 3.1 CONCLUSION FROM LITERATURE SURVEY

The literature survey concludes that IoT and automation technologies, especially those based on microcontrollers like ESP32, have significantly enhanced the efficiency, sustainability, and precision of hydroponic farming. The reviewed papers consistently demonstrate that integrating sensors for critical parameters such as pH, electrical conductivity, temperature, humidity, and nutrient concentrations which enables real-time monitoring and automated control of nutrient delivery, environmental conditions, and irrigation. This approach optimizes resource use, improves crop yield and quality, and reduces manual labor. Many systems utilize cloud platforms and mobile applications for remote data visualization and management, improving accessibility and scalability.

Methodologies vary, ranging from sensor networks with relay-based actuation to advanced implementations integrating machine learning algorithms for predictive analytics and automated decision-making. These systems have been applied in diverse environments, including vertical urban farms, indoor greenhouses, and aeroponic setups. Advantages highlighted include precise nutrient and pH management, energy and water conservation, and increased productivity. However, common limitations are sensor calibration drift, dependency on stable internet connectivity, high initial setup costs for complex systems, and challenges related to scaling and power efficiency.

Overall, while the integration of ESP32 and IoT in hydroponics represents a transformative advancement in smart agriculture, future research and development are needed to address scalability, data security, cost reduction, and robust long-term operation under varied conditions.

## 4. METHODOLOGY

The methodology of the smart hydroponic farming system project incorporates a structured approach to design, development, and implementation of automated environmental monitoring and control powered by the ESP32-S3 microcontroller. The system architecture consists of sensor modules measuring critical parameters such as pH, Electrical Conductivity (EC), Total Dissolved Solids (TDS), temperature, humidity, water level, and flow rate, all integrated with the ESP32-S3 for real-time data acquisition and processing. Actuators including nutrient dosing pumps, pH regulators, fans, and humidifiers are controlled via relay modules connected to the microcontroller.

Data collected by sensors are transmitted wirelessly via Wi-Fi using the ESP32-S3's built-in connectivity to the Blynk IoT platform, allowing remote monitoring and control from mobile devices. The architecture includes a feedback control loop where sensor data trigger automatic adjustments to maintain optimal growing conditions for the plants. This loop ensures precise environmental regulation with minimal human intervention.

The methodology includes component selection ensuring compatibility, power supply design for stable operation, and sensor calibration protocols to guarantee data accuracy. Software development on the ESP32-S3 involves writing efficient code to read sensor data, execute control logic for actuators, and manage wireless communication securely with the cloud platform. Calibration involves using standard buffer solutions for pH and EC sensors and testing sensor fidelity under variable environmental conditions.

Workflow proceeds through iterative testing and starting from single sensor validation, integration testing of multiple sensors and actuators, connectivity and cloud data visualization verification, followed by deployment in a controlled hydroponic setup for performance evaluation. The approach emphasizes modular design allowing future enhancements such as AI-driven predictive models and additional sensor integration.

This methodology ensures a robust, scalable, and reliable smart hydroponic system grounded in precision electronic control and IoT connectivity, aimed at maximizing crop yield, reducing waste, and enabling data-driven plant cultivation management.

## 4.1 ARCHITECTURE

The architecture depicted in the provided image represents a modern, IoT-based automated hydroponic farming system, leveraging the ESP32 microcontroller for centralized data acquisition, control, and cloud connectivity.

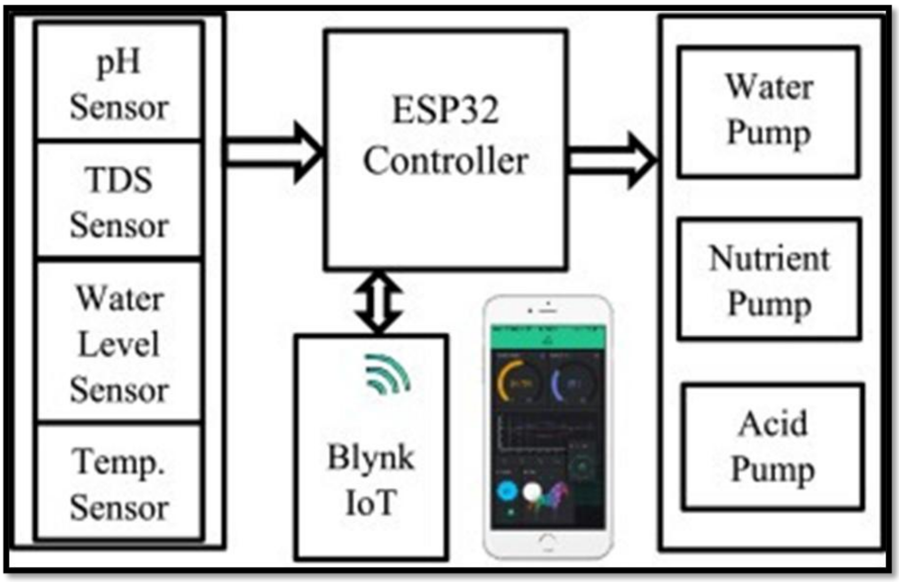


Fig1: Block Diagram

Detailed Architecture Description

1. Sensor Block:

This subsystem includes four core environmental sensors:

- pH Sensor: Continuously measures the pH value of the nutrient solution, ensuring it stays within a plant-appropriate range.
- TDS (Total Dissolved Solids) Sensor: Monitors the concentration of dissolved salts and fertilizers in the hydroponic solution, helping maintain optimal nutrient levels.
- Water Level Sensor: Prevents pump operation when water is low and enables automatic refilling or warnings.
- Temperature Sensor: Measures air or water temperature, which is critical for healthy root and foliage development.

Each sensor transmits its analog/digital readings to the ESP32 controller, typically via ADC channels or I2C/UART interfaces.

2. ESP32 Controller:

This Wi-Fi/Bluetooth-enabled microcontroller acts as the system's brain. It performs multiple roles:

- Collects and processes real-time data from all connected sensors.
- Compares sensed values against predefined thresholds ideal for plant growth.
- Executes embedded logic for automation, such as when to activate pumps and how much nutrient or acid to deliver.
- Maintains connectivity with the Blynk IoT platform for cloud dashboard integration, remote monitoring, and app-based commands.

3. Actuator Block:

Commands from the ESP32 controller drive key actuator devices:

- Water Pump: Manages the flow and circulation of the hydroponic solution, ensuring all plants are evenly watered.
- Nutrient Pump: Adds concentrated nutrient solution to maintain the correct TDS/EC in the plant reservoir.
- Acid Pump: Injects acid or base to correct pH as necessary for optimal nutrient uptake by plants.

Relays or MOSFETs isolate and safely power the high-voltage pumps guided by ESP32 signals.

4. IoT and Cloud User Interface:

Bidirectional communication with the Blynk IoT platform (over Wi-Fi) empowers users to:

- View sensor data, historical trends, and receive notifications on a smartphone or web dashboard.
- Execute manual overrides or adjust setpoints remotely for flexibility and fine-tuning.
- Receive real-time alerts in case parameters move outside optimal ranges, enabling rapid intervention.

5. System Workflow:

- Sensors constantly send raw data to the ESP32 controller.

- ESP32 process logic uses this data to automate pumps and maintain optimal hydroponic conditions—without human intervention.
- Sensor readings and control statuses are transmitted to the Blynk cloud, enabling visualization and manual interaction from anywhere with internet access.

Advantages of the Architecture:

- Offers continuous, precise environmental management for optimal plant growth.
- Reduces manual labor requirements and human error through intelligent automation.
- Supports real-time, remote monitoring, and control via smartphones.

Limitations:

- Relies on stable internet connectivity for full IoT functionality.
- Requires proper calibration and maintenance of sensors and pumps for sustained accuracy and effectiveness.

This architecture provides a robust, scalable solution suitable for hobbyist, educational, or commercial hydroponic operations. It blends embedded system design with modern cloud and mobile tech for next-generation agriculture.

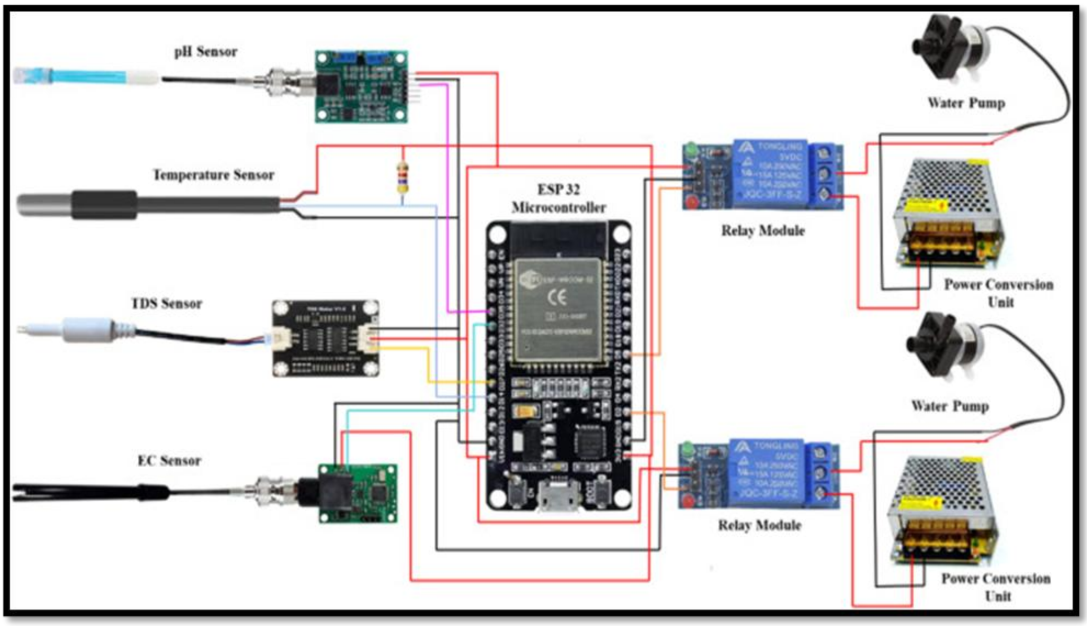


Fig. 2: Circuit Diagram

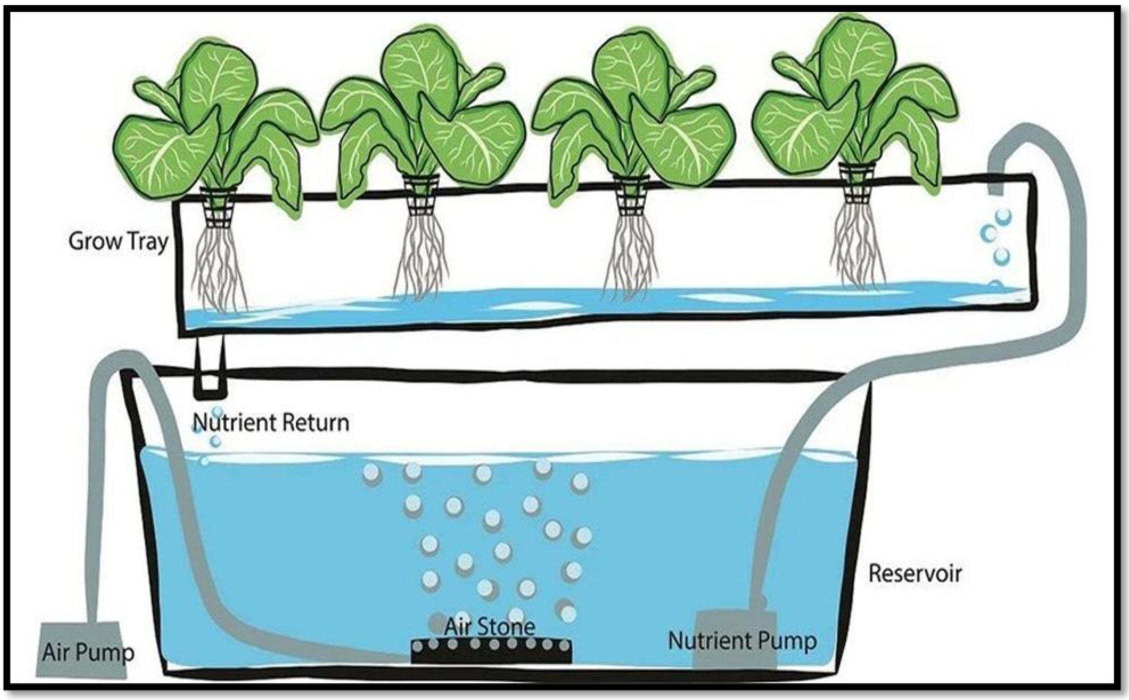


Fig. 3: Chassis

Chassis and Main Hardware Components

1. Grow Tray:
    - The upper horizontal container is called the grow tray, where plants are mounted.
    - The plants' roots are suspended through holes in the tray, allowing direct contact with the nutrient solution below.
    - The tray maintains a shallow layer of water so roots receive adequate water and nutrients while the upper portion of roots are exposed to air for oxygen uptake.
2. Nutrient Reservoir (Tank):
    - Located beneath the grow tray, the reservoir stores the bulk of the nutrient-rich water solution.
    - It serves as the central hub for nutrient, pH, and water level adjustments.
3. Water/Nutrient Circulation System:
    - Nutrient Pump: The pump moves the solution from the reservoir up into the grow tray in a controlled flow.
    - After passing over plant roots, excess solution drains back to the reservoir through a nutrient return line, creating a closed-loop system.

4. Aeration System:
   - Air Pump: Supplies air to the reservoir through an air stone placed at the bottom.
   - The air stone breaks the airflow into fine bubbles, increasing dissolved oxygen in the nutrient solution, which is vital for healthy root respiration.
5. Piping and Tubing:
   - Food-grade pipes or flexible tubes connect the water/nutrient pump to the tray and provide a return path.
   - Separate tubing connects the air pump to the air stone for aeration.

Functionality and Importance

- This structure ensures plants receive a steady supply of nutrients, water, and oxygen.
- Recirculation reduces water and fertilizer waste and keeps nutrients available to plant roots at all times.
- The modular nature of the chassis allows maintenance and scalable tray expansion for more plants.
- Key hardware namely the grow tray, reservoir, pumps, tubing, and aeration unit forms the foundation for adding sensors and connecting the whole system to a smart controller such as ESP32.

# 5. SYSTEM SPECIFICATIONS

## 5.1 HARDWARE SPECIFICATIONS

### I. ESP32- S3 Microcontroller:

Processor: Dual-core 32-bit Xtensa LX7 CPUs, running up to 240 MHz

Memory: 512 KB SRAM internal, 384 KB ROM, supports external PSRAM and Flash

Connectivity:

- Wi-Fi 802.11 b/g/n 2.4 GHz
- Bluetooth 5.0 (including BLE with longer range and improved throughput)

Peripherals:

- Up to 45 programmable GPIOs
- Multiple SPI, I2C, UART, PWM, ADC, DAC interfaces
- Supports USB OTG, LCD and camera interfaces (DVP)

Low Power: Includes ultra-low-power coprocessor, multiple sleep modes for energy-efficient operation

AI Accelerations: Added instructions optimized for machine learning workloads

Software Support: Compatible with Espressif's ESP-IDF development framework with rich libraries and tools

Fig. 4: ESP32 Microcontroller

## II. Sensors:

**A. pH Sensor**: A pH sensor, or pH electrode, is essential in hydroponic systems for monitoring the acidity or alkalinity of the nutrient solution, ensuring nutrient availability and optimal plant growth.

**Types and Range**

- Most hydroponic pH sensors are designed to measure from 0 to 14 pH.
- They can operate across a variety of temperatures (0°C to as high as 130°C for high-performance glass types).
- Materials can include glass (for high accuracy and harsh chemicals) or plastic/PTFE (for durability in field use).

**Calibration and Accuracy**

- pH sensors must be regularly calibrated, typically with standard buffer solutions (e.g., pH 4.00, 7.00, and 10.00).
- Multi-point calibration increases measurement accuracy, especially when the solution's pH may fall above or below neutral (7.00).
- Sensors drift over time due to electrode aging and fouling, requiring recalibration and maintenance for long-term accuracy.

**Features and Installation**

- Advanced sensors use solid dielectric and large PTFE liquid junctions for better life span and easier maintenance.
- Long reference diffusion paths and high-quality low-noise cables allow sensor output to travel long distances without interference.
- They may be installed via side wall, top flange, pipe, or submersible installation, with the interface ideally at a 15-degree oblique angle for best performance.

**Applications and Constraints**

- pH sensors are vital in hydroponics, industrial wastewater, biotech, food, pharmaceuticals, and more.

- Harsh chemicals, extreme temperatures, and physical fouling can reduce lifespan but robust sensor construction and regular calibration mitigate these effects.
- The practical response time for a good pH sensor is under a minute.

**Specifications:**

- Input Supply voltage:- (VDC) 5
- Module Size:- (mm) 50 x 47 x 16
- Measuring Range:- 0 to 14
- PH Measuring Temperature:- 0 to 50
- Accuracy:- 0.01 pH
- Response Time:- 1min
- Cable Length:- (cm) 75
- pH sensor size:- (mm) 150, 12

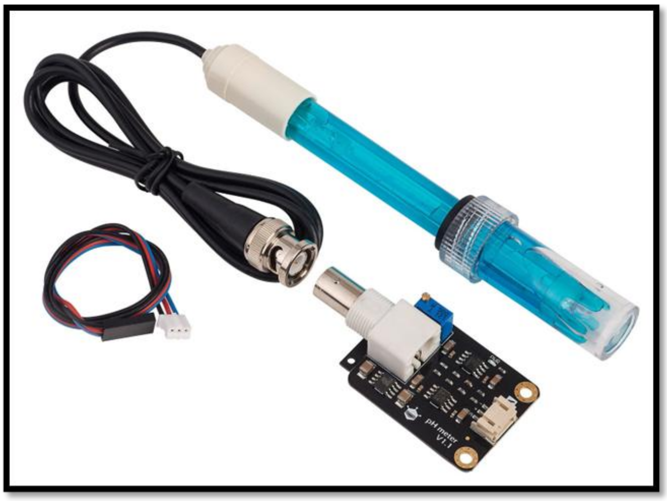

Fig. 5: pH Sensor

**B. Temperature & Humidity Sensor (DHT22 ):** Provides real-time data on air temperature and relative humidity within the grow area. The DHT22 is a widely used digital temperature and humidity sensor, well-suited for smart hydroponic systems, greenhouse environments, and indoor agriculture due to its precision, ease of interfacing, and reliability.

**Working Principle and Features**

- Sensor Structure: The DHT22 uses a capacitive polymer-based humidity sensor and a thermistor for temperature sensing, outputting a digital signal that can be directly read by microcontrollers like the ESP32.
- Output: Delivers both temperature and humidity readings via a single-wire digital interface (non-I2C, non-OneWire), reducing wiring complexity.
- Calibrated Signal: It comes pre-calibrated from the factory, with compensation coefficients stored in memory for precise readings and long-term repeatability.

**Specifications**

- Temperature Range: -40°C to +80°C
- Temperature Accuracy: ±0.5°C (typical)
- Temperature Resolution: 0.1°C
- Humidity Range: 0% to 100% RH
- Humidity Accuracy: ±2% RH (typical, max ±5% RH)
- Humidity Resolution: 0.1% RH
- Supply Voltage: 3.3V to 6V DC
- Sampling Rate: Once every 2 seconds
- Digital Output: Serial data, requires one GPIO pin for data line
- Dimensions: Typical body is 14x18x5.5 mm (small version) or 22x28x5 mm (large version)

**Hardware Integration & Usage**

- Connection: The DHT22 has four pins: Vcc (3.3-6V), Data (pull-up resistor required, usually 4.7-10kΩ), NC (not connected), and GND.
- Interfacing: Direct digital readout—libraries for Arduino/ESP32 are widely available. Each sensor should be connected to a unique data pin if using multiples.

**Advantages**

- High accuracy and broad sensing range make it apt for plant environmental monitoring.
- Low power consumption, enabling battery or solar operation in remote hydroponic setups.
- Off-the-shelf libraries and stable performance lead to quick and reliable integration.

**Limitations**

- Maximum sampling rate is once every 2 seconds—not suitable for very fast real-time control loops.
- Not waterproof; must be mounted away from direct water splash.
- Digital protocol requires precise timing, making careful coding necessary.


Fig:6 DHT22

C. **Water Level Sensor:** Detects nutrient solution/water levels in tank or channels. Maintains hydration; may trigger refilling. A water level sensor is a key component in hydroponic systems, ensuring that the nutrient solution reservoir maintains an optimal volume for continuous plant growth and preventing equipment damage or pump dry runs.

**Types and Working Principles**

Water level sensors in hydroponics come in several main types:

- Float Switch: A mechanical device where a buoyant float rises and falls with the liquid surface, triggering an electrical switch when a set level is reached. These are simple, low-cost, and suitable for discrete (high/low) level detection.
- Capacitive Sensor: Uses a pair of conductive electrodes. As water covers the probe, capacitance changes are measured, enabling continuous level monitoring. Good for non-contact measurement and for tanks of non-metallic construction.
- Ultrasonic Sensor: Emits ultrasonic pulses and measures the time taken for the echo to return from the water surface, calculating distance and thus the fluid level. These sensors are precise, non-contact, and unaffected by water chemistry, making them ideal for corrosion-prone environments.
- Radar Level Sensor: Uses microwave signals to measure the time delay from the sensor to the water surface. This advanced, non-contact method offers high precision and extreme reliability, even in the presence of vapours, mists, or corrosive chemicals, though at a higher cost.
- Optical Sensor: Uses an infrared (IR) light beam; changes in refraction when submerged vs. in air are detected.

**Features and Specifications**

- Accuracy: Ranges from ±2 mm for basic float switches to sub-millimetre for radar sensors.
- Materials: Sensors are available in chemically resistant plastics (e.g., PVC, PTFE, PVDF) for hydroponics, preventing sensor fouling or corrosion.
- Signal Output: Options include analog (0–5V, 4–20mA), digital (high/low), RS485 serial, and IoT-enabled protocols for cloud-based monitoring.

- Installation: Can be submersible or externally mounted (ultrasonic/radar/optical).
- Integration: Most water level sensors interface directly with microcontrollers like ESP32, providing real-time data for automated pump control or user alerts.

**Operation in Hydroponics**

- The sensor is typically placed within the reservoir or at critical points in the water flow.
- The system interprets sensor readings, triggering water pumps to refill or stop when the minimum or maximum setpoints are reached.
- Advanced systems use continuous sensors for proportional control, while simpler setups use high/low float switches for basic automation.

**Advantages:**

- Prevents pump dry runs, protecting equipment.
- Enables water-saving routines by maintaining minimal but sufficient volume.
- Supports remote data logging and automated control via IoT integration.

**Limitations:**

- Float switches can be affected by debris or root overgrowth.
- Capacitive and ultrasonic sensors may require calibration and periodic cleaning.
- High precision radar/ultrasonic versions have higher initial cost, but minimal maintenance.

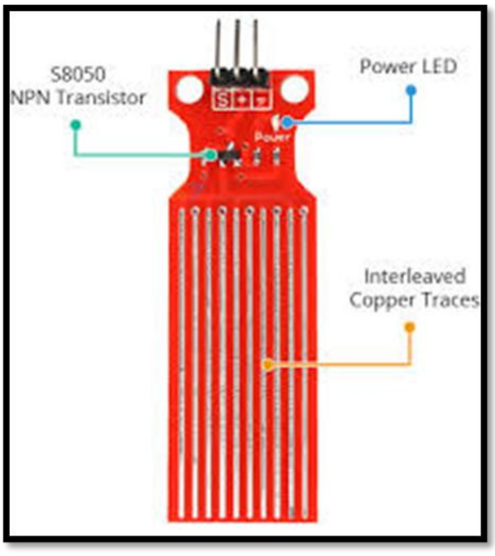


Fig. 7: Water Level Sensor

**D. Electrical Conductivity (EC)** sensors play a critical role in hydroponic systems by measuring the ability of the nutrient solution to conduct electrical current, which directly correlates to the concentration of dissolved salts—and hence, nutrients—available to plants. The EC value helps growers precisely control nutrient levels, avoiding deficiencies or toxicities that can negatively impact crop health and yield.

**Principle of Operation**

EC sensors typically consist of two or more electrodes submerged in the nutrient solution. An alternating current (AC) voltage is applied between the electrodes to prevent electrode polarization and corrosion. The sensor measures the conductance, or the ease with which electrical current passes through the fluid, which depends on ion concentration. The raw conductance value is then converted into conductivity units by accounting for the cell constant, related to electrode size and geometry.

**Key Specifications and Features**

- Measurement Range: Commonly from a few µS/cm to several mS/cm, suitable for typical hydroponic nutrient ranges.
- Accuracy: Typically, around ±1-2% of reading.
- Output: Analog voltage or digital output compatible with microcontrollers (e.g., ESP32) for real-time data acquisition.
- Frequency: Uses AC to actively measure to prevent electrode degradation.
- Auto-ranging: Some advanced EC sensors feature automatic range switching for accurate measurement across low and high conductivity values.

**Use in Hydroponics**

- By continuously monitoring EC, the hydroponic system can:
- Adjust nutrient dosing pumps to maintain ideal nutrient strength according to crop needs and growth phases.
- Detect dilution levels when water is added.
- Prevent nutrient toxicity caused by excessively high concentrations.

**Advantages**

- Provides quick, reliable feedback for nutrient management.
- Enhances crop yield quality and consistency by maintaining balanced nutrients.
- Reduces manual labor and chemical waste with automated dosing based on accurate measurements.
- Compatible with IoT microcontrollers enabling remote monitoring and control.

**Limitations and Considerations**

- Sensor fouling by mineral precipitates or biofilms can affect accuracy; regular cleaning and maintenance are necessary.
- Calibration is required for sustained accuracy, often using standard solutions.
- Sensor electrodes require replacement over time due to wear.

Fig. 8: EC Sensor

**E. TDS Sensor:** A Total Dissolved Solids (TDS) sensor is an important component in hydroponic systems that measures the concentration of dissolved solids mainly mineral salts and nutrients that are present in the water, which directly impacts the nutrient availability for plants.

**Principle of Operation**

TDS sensors estimate the total amount of dissolved solids by measuring the electrical conductivity (EC) of the solution, as dissolved salts increase the water's ability to conduct electricity. Unlike direct chemical analysis, TDS sensors provide a quick and indirect estimation of nutrient concentration, typically expressed in parts per million (ppm). They apply a small voltage through two electrodes immersed in the solution and measure the resulting electrical current to calculate conductivity, which is then converted to TDS.

**Specifications and Features**

- Measurement Range: Suitable for hydroponic nutrient concentrations, usually between 0 and 2000 ppm or higher.
- Accuracy: Approximately ±2% of reading.
- Output Signal: Analog voltage proportional to TDS or digital outputs for microcontroller compatibility.
- Response Time: Fast, typically under a few seconds.
- Calibration: Requires calibration with standard solutions similar to expected nutrient concentrations.
- Maintenance: Regular cleaning recommended to avoid fouling or coating on sensor probes.

**Application in Hydroponics**

Monitoring TDS is crucial for maintaining the correct balance of nutrients required by plants during various growth stages. The sensor helps growers:

- Detect nutrient deficiencies or toxicities early.
- Adjust nutrient dosing precisely for optimal growth and yield.
- Monitor changes due to dilution, evaporation, or plant uptake.
- Maintain consistent solution strength to prevent stunted growth or nutrient burn.

**Advantages**

- Simple, fast, and cost-effective way to estimate dissolved nutrient concentration.
- Easily interfaceable with microcontrollers like ESP32 for automation.
- Enhances system control, improving crop health and reducing waste.

**Limitations**

- TDS is an indirect measure and cannot identify specific ions or contaminants.
- Sensor readings can be affected by temperature; thus, temperature compensation is often needed.
- Regular calibration and cleaning are necessary for accuracy and longevity.

Fig. 9: TDS Sensor

### III. Power Supply:

A 12V DC Li-ion battery is a popular and efficient power supply option for hydroponic and embedded system

**Basic Specifications**

- Voltage: Typically 12V nominal, composed of 3 or 4 lithium-ion cells (each ~3.7V nominal) connected in series.
- Capacity: Common capacities range from 7Ah to 12Ah or more, depending on the cell configuration, determining runtime.
- Energy: For example, a 12V 12Ah battery stores about 144Wh of energy (12V × 12Ah).

**Features and Advantages**

- Lightweight and Compact: Li-ion batteries offer high energy density, making them lighter and smaller than equivalent lead-acid batteries, enhancing portability.
- Rechargeable: They typically allow hundreds to thousands of charge-discharge cycles, extending operational life versus disposable batteries.
- Built-in Protection (BMS): Most packs include a Battery Management System that guards against overcharge, deep discharge, short circuits, and temperature extremes, ensuring safe operation.
- Fast Charging: Charging times generally range from 2 to 5 hours with appropriate chargers.
- Stable Output: Provides consistent 12V output, essential for stable motor, sensor, and microcontroller operation.

**Application in Hydroponics**

- Reliable Power Source: Supplies continuous power to Arduino/ESP32 controllers, motors/pumps, sensors, and actuators.
- Portability: Enables off-grid or mobile hydroponic setups, making the system independent of fixed power outlets.
- Energy Efficiency: Supports integration with solar or other renewable energy kits for sustainable farm operation.
- Backup Power: Acts as a battery backup during power outages, preventing abrupt interruption in system control that could harm plants.

**Considerations**

- Charging Requirements: Requires a proper Li-ion charger matching voltage and current specifications to ensure battery longevity.
- Environmental Sensitivity: Must be used within specified temperature ranges to avoid damage or reduced capacity.

- Disposal: Requires proper recycling or disposal following environmental safety norms.

**Example Specs (from typical commercial packs)**

- Voltage: 12.6V max charged, 12V nominal
- Capacity: 7.8Ah to 12Ah
- Weight: ~0.9 to 1.2 kg
- Protection Circuit: Overcharge, over-discharge, short circuit, temperature
- Lifetime: 1000+ full charge cycles
- Charging Time: 3–5 hours (depending on charger)

## 5.2 SOFTWARE SPECIFICATIONS

### I. Arduino IDE:

Arduino IDE (Integrated Development Environment) is a free, open-source software platform designed to write, compile, and upload code to microcontrollers like Arduino boards and ESP32 modules, making it immensely popular among hobbyists, students, and professionals for embedded system development.

**Features and Components**

- Code Editor: A simple, user-friendly text editor with syntax highlighting, code completion, and bracket matching to write Arduino programs (sketches) in C/C++.
- Compiler: Translates human-readable Arduino code into machine code that microcontrollers execute. It uses the GCC toolchain behind the scenes.
- Uploader: Communicates with Arduino/ESP boards over serial or USB to flash compiled code onto the device.
- Serial Monitor: Allows runtime interaction with the microcontroller via serial input/output for debugging and data viewing.
- Library Manager: Manages external code libraries for sensors, actuators, and communication protocols, facilitating easy integration.
- Board Manager: Supports dozens of microcontrollers including various Arduino boards and ESP32/ESP8266 devices by installing board definitions and drivers.

**Working with Hydroponic Systems**

- Arduino IDE supports rapid prototyping of sensor integration (pH, EC, TDS, DHT22, water level), actuator control (pumps, relays), and IoT communication (Wi-Fi with ESP32).
- Supports async and timer-based control necessary for real-time environment management in hydroponics.
- Provides extensive libraries like Blynk for IoT dashboards, OneWire for DS18B20 temperature sensors, and sensor-specific libraries easing hardware interfacing.

**Advantages**

- Straightforward setup and cross-platform (Windows, macOS, Linux).

- Large community with abundant tutorials and example codes.
- Modular and extensible for diverse projects from simple to advanced automation.
- Supports debugging via serial print statements and on-chip debugging tools.

**Limitations**

- May require manual management of dependencies for complex projects.
- Limited native debugging features compared to professional IDEs without advanced tooling.
- Tick-based loop model may require careful timing code for highly concurrent tasks.

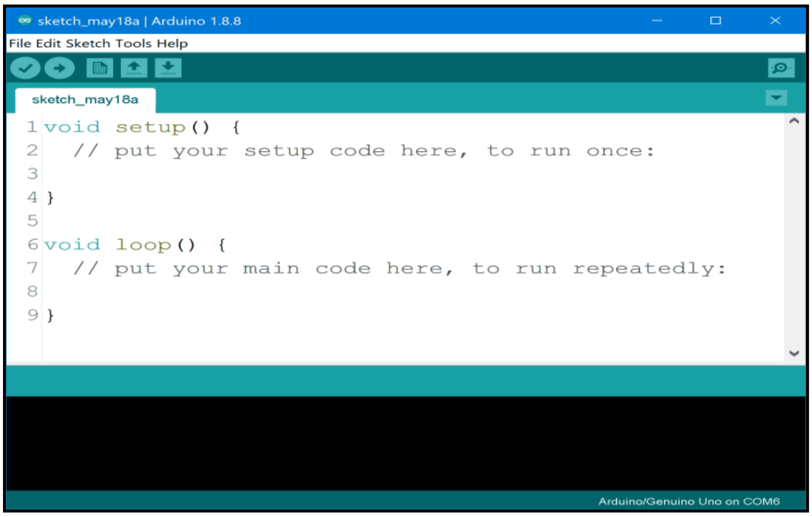


Fig. 10: Arduino IDE code window

Fig. 11: Arduino IDE logo

### II. Proteus 8 Professional

Proteus 8 Professional is an advanced electronic design automation (EDA) software widely used for designing, simulating, and testing electronic circuits and printed circuit boards (PCBs). Developed by Labcenter Electronics, this professional suite combines schematic capture, circuit simulation, PCB layout, and microcontroller firmware simulation into a single integrated environment, making it an invaluable tool for engineers, educators, and hobbyists.

**Key Features:**

- Schematic Capture: Users can create detailed electronic circuit diagrams with a comprehensive library of components, including resistors, capacitors, semiconductors, microcontrollers (PIC, AVR, ARM), sensors, and modules. Custom components can also be modelled and added.

- Circuit Simulation: Proteus supports both analog and digital circuit simulation using SPICE engine for accurate analog behaviour and virtual instruments (oscilloscope, logic analyser, signal generator) for interactive testing. Real-time simulation helps debug and optimize circuits before hardware implementation.

- Microcontroller Simulation: A unique strength of Proteus 8 Professional is its ability to simulate microcontroller firmware directly within the circuit, enabling complete embedded system testing including code execution, IO interaction, and peripheral operation, without physical hardware.

- PCB Design and Layout: Proteus integrates powerful tools for designing PCB layouts, including multi-layer boards, trace routing, auto-router, footprint libraries, and 3D board visualization. It facilitates transition from schematic to manufacturing-ready layouts.

- User Interface: The software offers an intuitive drag-and-drop environment, real-time design rule checks, multi-window views, and extensive documentation and tutorial support.

- Code Debugging: Provides debugging tools such as real-time variable monitoring, breakpoints, and step-through code execution within the microcontroller simulation context.

- Component Libraries: Extended libraries cover thousands of components with electrical and physical properties, making it easy to simulate realistic circuits.

- Compatibility and Integration: Supports Windows platforms (7, 8, 10, 11), with integration options for external code editors and CAD import/export.

**Applications:**

Proteus 8 Professional is extensively used in academia for teaching electronics and embedded systems, in industry for prototype circuit verification and product development, and by researchers and hobbyists for design automation and testing.

**Advantages:**

- Comprehensive and unified design/simulation flow ensures early detection of design flaws.
- Accurate microcontroller firmware simulation reduces dependency on physical prototypes.
- 3D visualization aids in understanding and correcting physical layout issues.
- Large component databases and community support.

**Limitations:**

- Licensing cost can be significant for individual hobbyists.
- The learning curve may be steep for beginners without prior EDA experience.

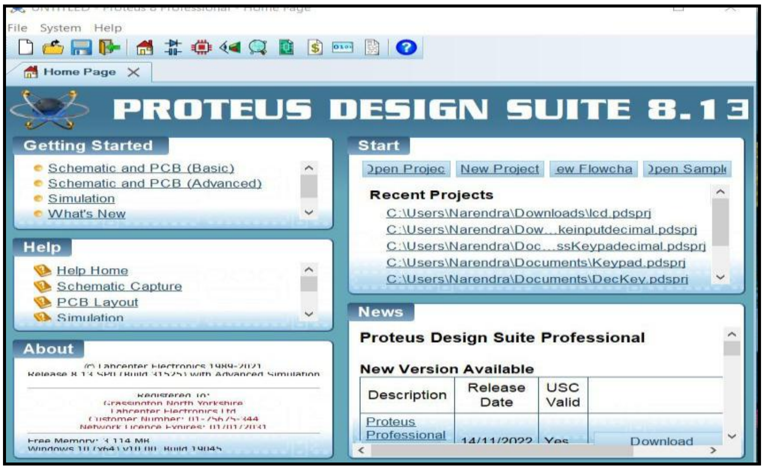


Fig. 12: Proteus 8 Professional

### III. Blynk IoT

Blynk IoT is a powerful and user-friendly Internet of Things platform designed to connect, control, and manage smart devices remotely through the cloud. It enables rapid IoT project development with minimal coding by providing a low-code environment with drag-and-drop app building, cloud connectivity, and device management.

**Core Features:**

- Mobile and Web Dashboards: Blynk offers customizable dashboards on mobile (iOS and Android) and web platforms for real-time visualization of sensor data and device status.
- Drag-and-Drop Interface: Users can create custom IoT apps using widgets like buttons, sliders, gauges, and charts without coding, allowing easy manual or automatic control over connected devices.
- IoT Device Management: Blynk supports provisioning, profiling, and secure management of many devices simultaneously, including role-based access control for multi-user environments.
- Real-Time Monitoring and Control: Devices communicate with the Blynk cloud via Wi-Fi, cellular, or other protocols. Bidirectional data flow enables live sensor updates and remote actuation.
- Cloud Infrastructure: Fully hosted, scalable cloud backend handles data storage, analytics, notifications, and firmware over-the-air (OTA) updates.
- Third-Party Integrations: Supports integration with Alexa, Google Assistant, IFTTT, REST APIs, enabling expanded IoT workflows.
- Firmware Library: Blynk provides firmware libraries for microcontrollers (Arduino, ESP32, Raspberry Pi, STM32, etc.) to simplify connection and communication with the platform.

**Architecture:**

- Blynk App: User interface for device control, data visualization, and alerting.
- Blynk Server: Cloud or local server mediates communication between apps and devices.
- Device Firmware: Client-side code (running on ESP32 or other MCUs) manages sensors and actuators and handles cloud communication.

**Benefits:**

- Fast project deployment and prototyping.
- No need for complex backend or mobile app programming.
- Secure data management and granular sharing controls.
- Large developer community and extensive documentation.

**Use in Hydroponics:**

Blynk IoT enables real-time remote monitoring of sensor data (pH, temperature, EC, TDS, water level) and remote control of pumps and actuators in hydroponic systems. Users receive instant alerts and can adjust system parameters from anywhere, improving precision agriculture and resource efficiency.

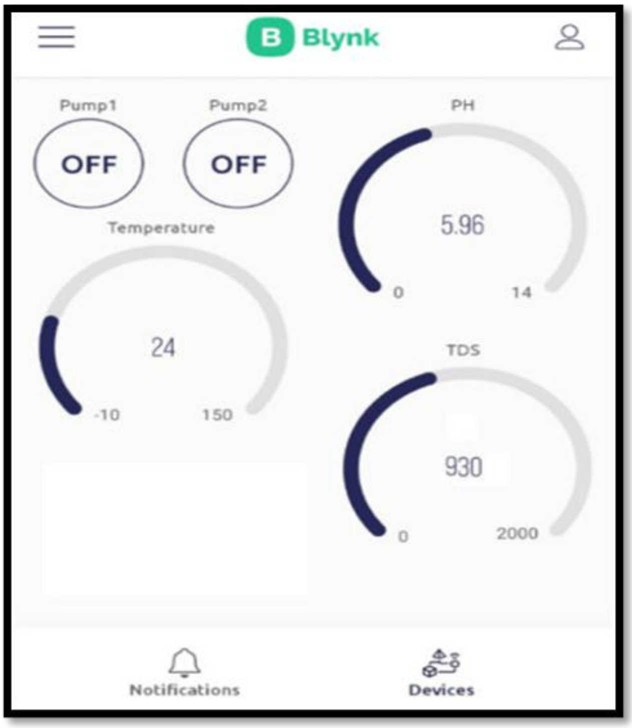


Fig.13: Blynk IoT Mobile setup

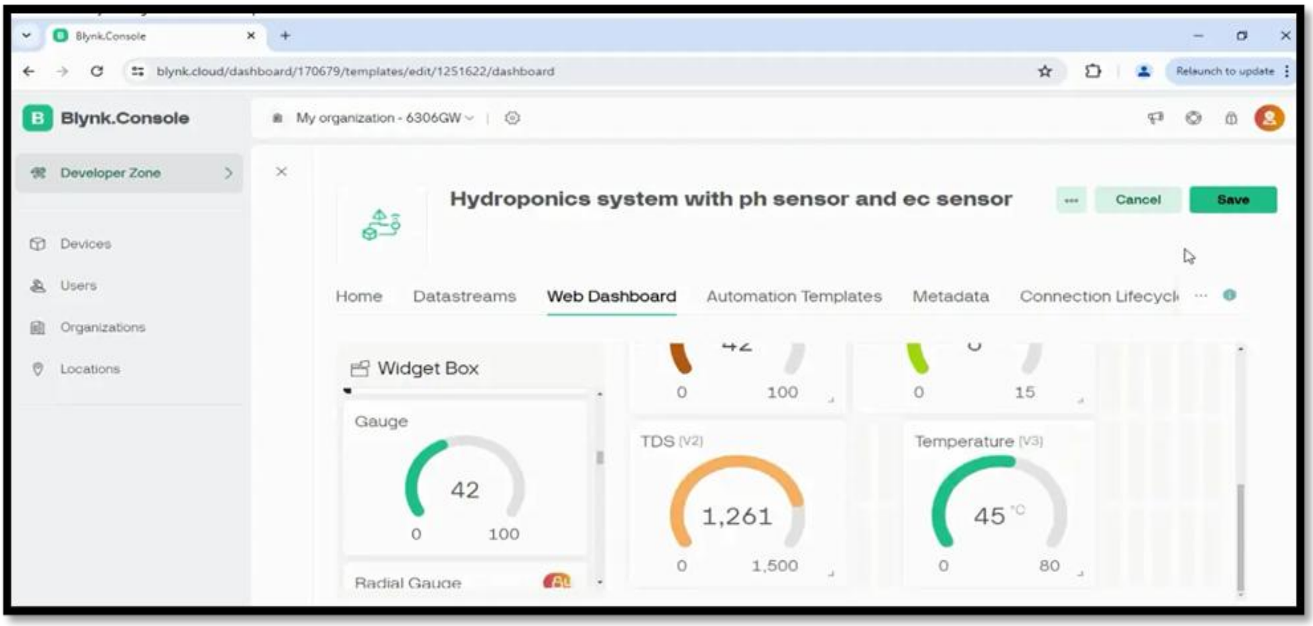


Fig. 14: Blynk IoT Dashboard

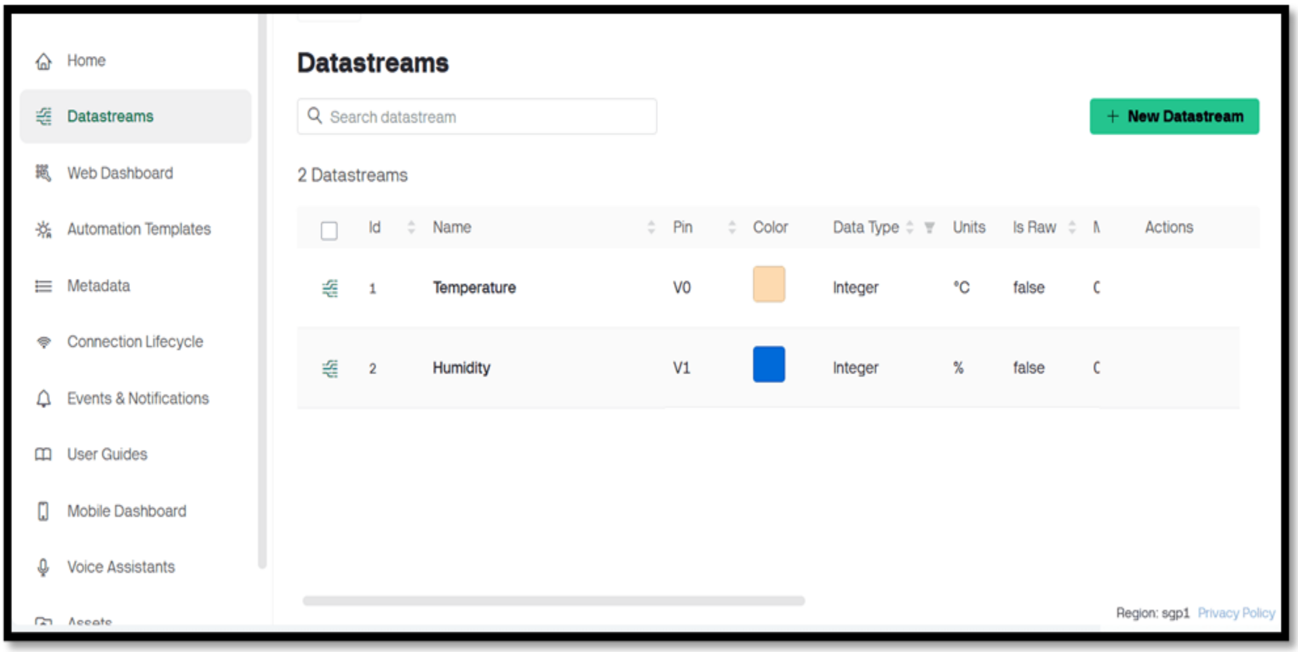


Fig. 15: Datastreams

# 6. RESULT

## 6.1 PLANTS CONSIDERED

- Many different flowers, vegetables and plants can be grown hydroponically.
- The fruits and vegetables that can be grown include Leafy greens like lettuce and spinach.
- Tomatoes, Beans and herbs include basil, mint and Peppers, ginger and okra.

| Plant Name | Temperature Range (°C) | Humidity Range (%) | pH Range | Approx. Growth Time (Days) |
|---|---|---|---|---|
| Tomato | 21 - 27 | 50 - 70 | 6.0 - 6.5 | 60+ (not under 30 days) |
| Basil | 20 - 30 | 40 - 70 | 5.5 - 6.5 | 25 - 30 |
| Mint | 18 - 24 | 40 - 70 | 5.5 - 6.5 | ~30 |
| Radish | 16 - 24 | 60 - 70 | 6.0 - 7.0 | 20 - 30 |
| Spinach | 15 - 24 | 60 - 70 | 5.5 - 6.8 | 25 - 30 |
| Green Beans | 18 - 27 | 50 - 70 | 6.0 - 6.5 | 30 - 40 |
| Lettuce | 18 - 24 | 50 - 70 | 5.5 - 6.5 | 30 |
| Peppers | 21 - 26 | 50 - 70 | 5.5 - 6.5 | 60+ (longer cycle) |

Table 1: Favorable Conditions

## 6.2SIMULATION

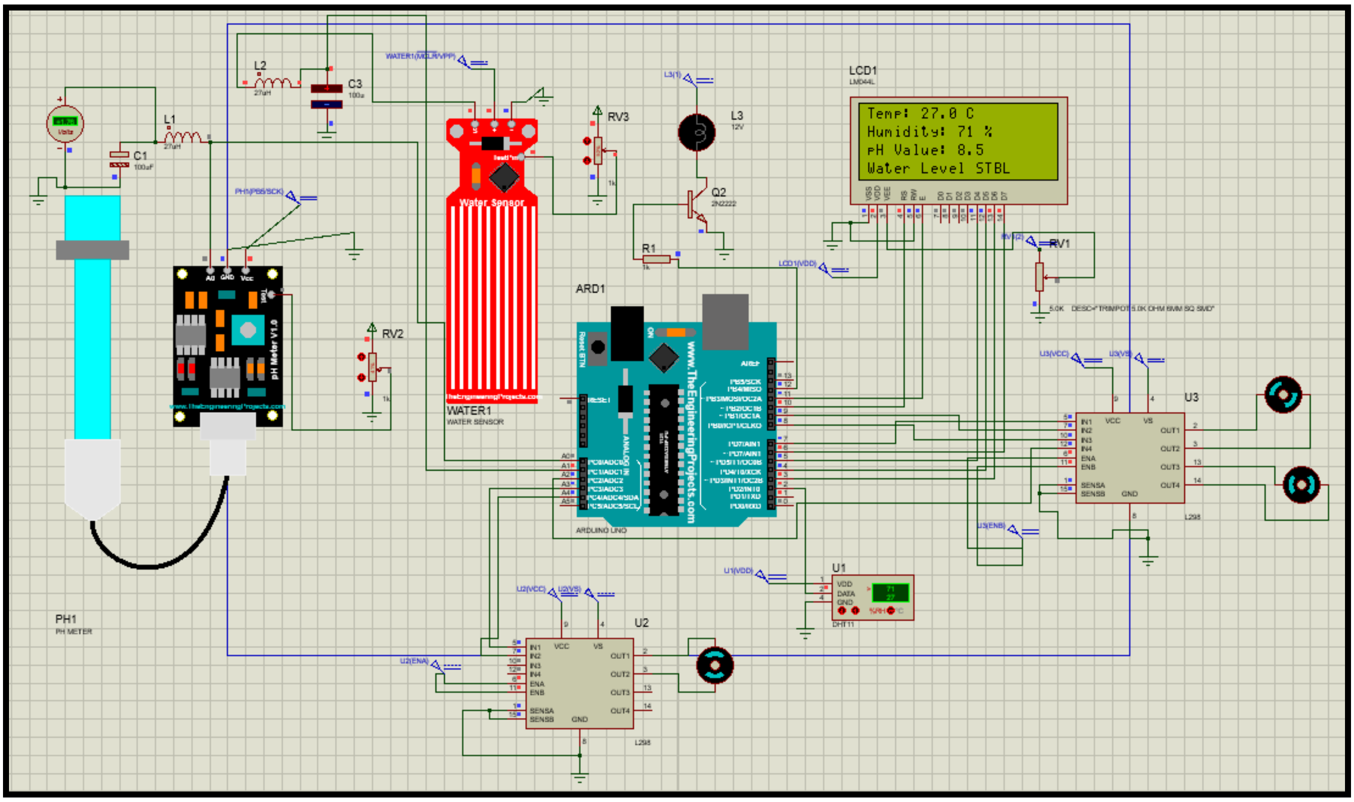


Fig.16: Proteus Simulation

This simulation, run in Proteus 8 Professional, models an automated hydroponic control system integrating real-world sensors, a microcontroller, actuator drivers, and a display interface to mimic practical hardware interactions. In this simulation setup using Proteus 8 Professional, the hydroponic control system is designed with two motors (acting as pumps), a lamp load, and temperature and water level regulation logic integrated through an Arduino Uno.

**Detailed Logic and Simulation Explanation**

**Pump (Motor) Control via Water Level Sensor:**

- The water sensor module detects the presence and level of water in the reservoir or grow tray.
- When water is sensed (level reaches a set threshold indicating the tank is not empty), the Arduino's control logic processes the sensor input.
- The microcontroller then sends output signals to activate one or both motors through an L298 motor driver IC. These motors mimic the action of water/nutrient pumps, ensuring solution circulation or refilling.
- If the water sensor detects insufficient water, the pumps remain off, protecting against dry runs.

**Lamp Load as Heating Element:**

- The simulation includes a lamp which acts as a heating device (analogous to a real-world heater lamp).
- The Arduino monitors temperature readings from the DHT11/DHT22 sensor.
- If the measured temperature is below the desired threshold (25–26°C), the Arduino energizes the lamp via a transistor switch (acting as a relay), thus heating the environment or nutrient solution.
- This logic enables the system to automatically maintain a minimum safe temperature for optimal root and plant development.

**Fan Representation by Motor (Ventilation Control):**

- The system also uses a motor (controlled through the driver IC) to mimic a ventilation fan.
- When the temperature exceeds a higher threshold (e.g., above 30°C), the Arduino logic triggers the motor to turn on, representing fan activation to dissipate excess heat and lower the temperature.
- This prevents overheating, which could otherwise stress or damage the plants.

**Display and Feedback:**

- The LCD module provides real-time display of sensor data: current temperature, humidity, pH value, and water level status. States like "lamp ON" or "fan ON" can also be shown as feedback.
- The coordinated response between sensed environmental parameters (water and temperature) and the action of hardware elements illustrates a practical automation routine for hydroponics.

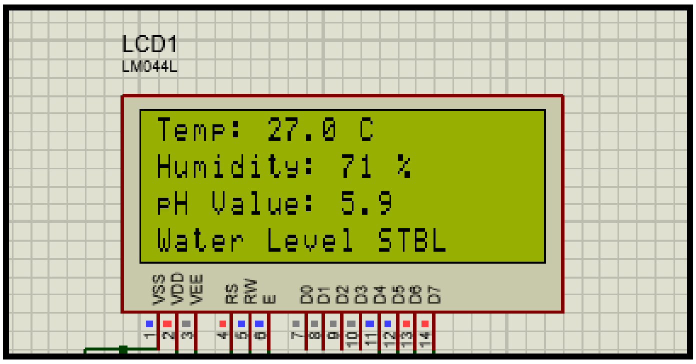


Fig 17: LCD Output

## 6.3 RESULT

The simulation of the intelligent hydroponic farming system has been successfully completed, confirming stability and optimal configuration of plant environments using selected sensor types and variable inputs. The structural design for the hydroponic chassis has been finalized after evaluating multiple configurations; the chosen structure meets criteria for durability, ease of sensor integration, and accessibility for maintenance.

. The team has successfully shortlisted a full suite of environmental sensors, including pH, temperature, humidity, electrical conductivity (EC), water level, and light sensors, carefully selected to match the agronomic requirements of various herbs, leafy greens, and vegetable crops. These selections are based on validated plant growth parameters and insights gained from extensive research literature to ensure optimal growth conditions. Sponsorship has been secured, ensuring supply of high-quality sensors and essential hardware; this partnership has enabled the team to move from simulation to physical prototyping with robust industry support.

# 7. COMPARATIVE ANALYSIS

| Aspect | Traditional Hydroponics | Automated (Basic) Hydroponics | Smart IoT-Based System (This Project) |
|---|---|---|---|
| **Nutrient Management** | Manual mixing/monitoring | Timer-based dosing; no sensor feedback | Automated EC/TDS & pH-controlled dosing, sensor-driven |
| **Water Management** | Manual refill | Float switches for automatic refill | Water level sensors, real-time status & pump actuation |
| **Temperature Control** | Manual (user checks/switches heater/fan) | Simple bi-metallic thermostat | DHT sensor monitors, MCU toggles lamp/fan on logic basis |
| **Monitoring** | Physical check only | Indicator LEDs or buzzers | Real-time LCD + mobile app with Blynk IoT, historical data |
| **User Intervention** | High | Medium | Very low; only for alerts/maintenance |
| **Customization** | Difficult: hardware change | Possible, but logic rewiring | Flexible via code/dashboard, easy scaling |
| **Data Logging/History** | None | Minimal (e.g., max/min indicators) | Complete cloud analytics and history |
| **Alerts/Remote Control** | None | No | Instant push notifications, remote override |
| **Resource Efficiency** | Low (risk of under/over dosing) | Moderate | High, precise, data-driven |
| **Deployment Cost** | Lowest | Moderate | Highest (more sensors, IoT modules), offset by ROI |

Table 2: Comparative Analysis

# 8. MERITS AND DEMERITS , APPLICATIONS

## 8.1 MERITS AND DEMERITS

**Merits:**

- Automated Plant Care: The system automates critical environmental and nutrient parameters (pH, EC/TDS, water level, temperature, humidity), drastically reducing manual effort and the risk of human error.
- Precision and Resource Optimization: Real-time sensor feedback ensures only the required amount of water and nutrients are supplied, minimizing waste and maximizing crop yield and health.
- Remote Monitoring and Control: Integration with IoT platforms (such as Blynk) allows users to monitor system status, receive alerts, and control actuators from anywhere using a smartphone or web dashboard.
- Scalability and Modularity: The system's modular architecture makes it easy to expand: more sensors, pumps, or control logic can be added with minimal redesign, supporting different plant types or larger farms.
- Data Logging and Analytics: Continuous data collection enables tracking trends, troubleshooting issues, and optimizing growing conditions for future cycles.
- Research & Educational Value: The system is a practical platform to teach and learn about microcontroller programming, sensor integration, automation, and IoT concepts.
- Flexibility: Control thresholds, responses, and behaviors can be updated via software, enabling customization for different seasons, crops, or operational preferences.

**Demerits:**

- Initial Cost and Complexity: High-precision sensors, actuator drivers, and IoT modules increase initial costs and project complexity compared to simple or traditional hydroponic approaches.
- Maintenance: Sensors (especially pH, EC, water level) require regular cleaning and periodic calibration to maintain accuracy, adding to ongoing maintenance requirements.
- Power Supply Dependencies: The system's operation is dependent on continuous power; outages may disrupt environmental management unless backups are in place.

## 8.2 APPLICATIONS

The applications of this ESP32- and IoT-based automated hydroponic system project in detail include:

- Commercial Hydroponic Farming: The system can be deployed in large-scale hydroponic farms to automate and optimize nutrient and environment control, improving crop yield and quality while minimizing labor and resource wastage. Real-time IoT monitoring facilitates centralized farm management and predictive maintenance.
- Urban and Vertical Farming: Ideal for space-conscious urban farms and vertical agriculture systems where precision control helps sustain plants in controlled environments at home, on rooftops, or in urban buildings. The mobile and cloud app interface simplifies remote management for urban growers.
- Research and Academic Use: This system serves as a versatile platform for students and researchers to study plant growth under controlled environmental parameters with precise sensor feedback. Data logging capabilities support scientific experimentation and advanced agronomic studies.
- Specialty Crop Production: Enables cultivation of medicinal herbs, high-value specialty plants, or tissue cultures requiring strict control of pH, nutrient concentration, water availability, and temperature, thereby supporting niche agricultural markets.
- Agricultural Training and Demo: Functions as an educational tool to demonstrate the integration of microcontrollers, sensors, and IoT in modern smart farming practices, boosting digital agriculture literacy.
- Food Security and Disaster Relief: Portable, automated hydroponic kits based on this design could be used to grow fresh food in remote or disaster-stricken areas with minimal human intervention.
- Home Automation and Hobby Farming: Enables indoor or greenhouse growers to maintain healthy vegetable/herb gardens with automated water and nutrient regimens and remote smartphone monitoring, fostering sustainable personal food production.
- Smart City Agriculture Integration: The IoT connectivity and cloud-based management support integration with broader smart city initiatives aimed at sustainable food systems, resource efficiency, and urban green infrastructure development.

## 9. CONCLUSION

The project demonstrates the simulation design of a smart hydroponic system employing the ESP32 microcontroller, advanced sensors (pH, EC/TDS, temperature/humidity, water level), and IoT-based remote monitoring via the Blynk platform.

Simulation results validated the sensor-actuator interaction model, confirming that motorized pumps respond accurately to water level inputs, while thermal regulation is effectively managed by controlling lamps and fans based on temperature thresholds. Real-time data display on the LCD and cloud dashboards provided comprehensive system visibility.

This automation minimizes manual labor, prevents human errors, and optimizes resource usage in hydroponic farming, contributing to enhanced crop yield, consistent quality, and sustainable production cycles. The IoT integration offers valuable flexibility and scalability for remote monitoring and control, essential for modern urban agriculture.

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